

TRIPLYTICS CHALLENGE

Build a Statistical Model and Framework to Generate District Level Market Attractiveness Index (Indian Pharmaceutical Industry) scores for over 700+ districts in India



ANALYZE

Uncover Key
Market Drivers



INSIGHT

Transform Data into
Actionable Insights



STRATEGIZE

Prioritize. Plan.
Create Impact.

A DATA DRIVEN APPROACH TO
POWER THE FUTURE OF PHARMA
GROWTH



700+
DISTRICTS



MULTI-DIMENSIONAL
DATA FRAMEWORK



STRATEGIC IMPACT
AT DISTRICT LEVEL

From Guesswork to Granular Intelligence:

A Predictive District Level Market Attractiveness Framework



THE PROBLEM

Today's district prioritization relies on competition benchmarking & doctor coverage missing disease burden, affordability, and future demand shifts



OUR SOLUTION: 3-INDEX MARKET ATTRACTIVENESS FRAMEWORK

Holistic view of district attractiveness across all key dimensions

Captures long term disease burden, patient pool stability & affordability dynamics

Captures seasonal / episodic demand, accessibility & infrastructure enablers



One Framework. Three Perspectives. Granular Intelligence.
District level scores tailored for strategic decision making.

WHAT MAKES IT DIFFERENT



Combines 8 Dimensions:
macroeconomic, epidemiological, patient pool, infrastructure, affordability, provider density, leading indicators, +1 Unique Variable: Mobility & Access Friction Index



Current + Forward looking view:
Predicts future high growth districts, not just a snapshot.



Built on 700+ districts, fully replicable & scalable: Across therapies and geographies.



Statistically rigorous: Weighted composite scoring validated against real district level proxies & outcomes.

BUSINESS IMPACT



SALES FORCE DEPLOYMENT
Right People in the right districts



MARKET PRIORITIZATION
Focus resources on highest potential districts



TERRITORY PLANNING
Design winning territories for sustainable growth



Turning 700+ districts
into a ranked, actionable roadmap for growth.

Decoding the Index: What Each Score Means & How It Drives Decisions

Three Lenses. One District. Complete Business Context.



**WHAT IT CAPTURES**

Blends macroeconomic strength, disease burden, access, and affordability into one opportunity score.

**HOW TO USE IT**

Broad market entry decisions, HQ-level resource allocation, and new district launches.

**SUMMATION**

Overall MAI = $w_1(\text{Macroeconomic}) + w_2(\text{Disease Burden}) + w_3(\text{Access}) + w_4(\text{Affordability}) + w_5(\text{Leading Indicators})$ *Weights derived via PCA + Expert Calibration*



**WHAT IT CAPTURES**

Long-term disease prevalence, patient pool stability, and affordability of sustained treatment.

**HOW TO USE IT**

Chronic portfolio expansion, adherence program targeting, and long-term rep assignment in stable markets.

**SUMMATION**

Chronic MAI = $w_1(\text{Chronic Disease Burden}) + w_2(\text{Patient Pool Stability}) + w_3(\text{Affordability}) + w_4(\text{Repeat Access Infra})$ *Weights derived via PCA + Expert Calibration*



**WHAT IT CAPTURES**

Seasonal illness patterns, immediate accessibility, and infrastructure responsiveness.

**HOW TO USE IT**

Seasonal campaign planning, rapid response distribution, and short-term rep surges during outbreak seasons.

**SUMMATION**

Acute MAI = $w_1(\text{Seasonal Incidence}) + w_2(\text{Immediate Access}) + w_3(\text{Infra Responsiveness}) + w_4(\text{Short-cycle Demand})$ *Weights derived via PCA + Expert Calibration*

HOW THEY DIFFER AT A GLANCE		OVERALL MAI	CHRONIC MAI	ACUTE MAI
   	Time Horizon	Long + Short-term blend	Long-term (repeat demand)	Short-term (episodic)
	Core Driver	Composite, all dimensions	Disease burden + affordability	Access + responsiveness
	Best For	Strategic entry decisions	Rep retention & loyalty markets	Seasonal surge planning

What We Measured and Why: Variables Behind The Score

Eight Dimensions. Real Data. No Guesswork.



Sr. No	DIMENSIONS		KEY VARIABLE(S).	DATA SOURCE	OVERALL / CHRONIC / ACUTE WEIGHTING
1		Macro-economic	District GDP proxy, per-capita income, urbanization %	 Census 2011 + State Economic Surveys	 All three (base layer)
2		Disease Burden	NCD prevalence rate (diabetes, CVD, hypertension)	 ICMR-NCD Reports,NFHS-5	 Chronic-heavy
3		Epidemiology (Acute)	Seasonal infection incidence (dengue, malaria, respiratory infections)	 NVBDCP,IDSP data	 Acute-heavy
4		Patient Pool & Access	Patient-to-doctor ratio, population density	 NHP (National Health Profile), Census	 All three
5		Provider Availability	Doctor density, hospital bed count	 NHP, State Health Directorates	 All three
6		Healthcare Infrastructure	PHCs/CHCs per capita, pharmacy density	 Rural Health Statistics	 Acute-heavy
7		Affordability & Spending	Out of pocket health expenditure, insurance penetration	 NSSO, NFHS-5	 Chronic-heavy
8		Leading Indicators	Income growth rate, urbanization trend, literacy trend	 State Economic Survey (3 year trend)	 All three



**+1 UNIQUE VARIABLE :
MOBILITY & ACCESS
FRICTION INDEX**



A composite proxy measuring the real world effort a patient needs to reach a pharmacy/provider combining road connectivity (from OSM/State PWD data), distance to nearest PHC, and public transport density.

Captures "invisible" access gaps that pure doctor density metrics miss especially relevant in semi-urban and rural high growth districts.



Two districts can have identical doctor density but very different real accessibility this variable exposes that gap.



CHRONIC THERAPY PRIORITIES:

Disease persistence, affordability of sustained treatment, repeat access infrastructure



ACUTE THERAPY PRIORITIES:

Seasonal incidence spikes, immediate accessibility, rapid infrastructure response

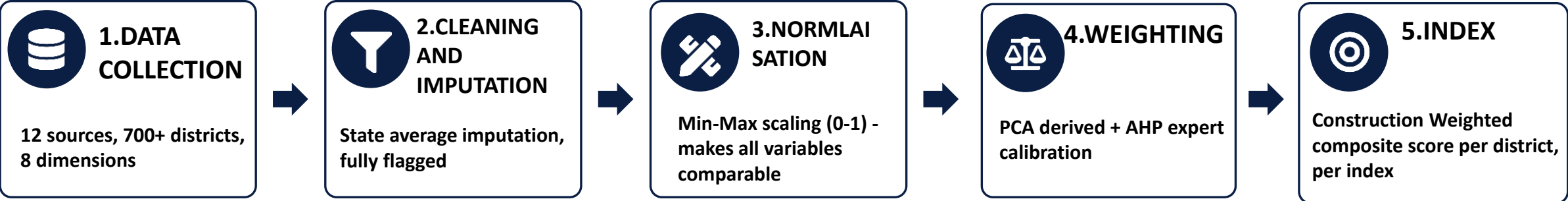


SAME 8 DIMENSIONS,DIFFERENT LENS:

Because chronic and acute markets behave fundamentally differently.

Building the Score: Data to Decision, Step by Step

A Predictive District Level Market Attractiveness Framework



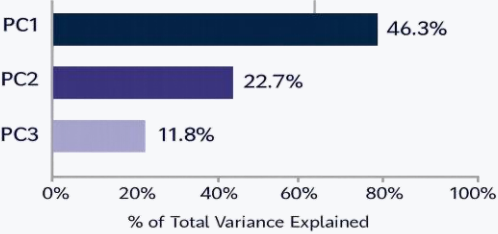
WEIGHTING: HOW WE GOT FROM 8 VARIABLES TO 1 SCORE

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A. PCA (Statistical Layer)

Principal Component Analysis run on all 8 dimensions to identify which variables explain the most variance in district level pharma opportunity removes redundancy, avoids double-counting correlated variables (e.g., income & urbanization).

Explained Variance by Principal Components
(Illustrative – top 3 components explain ~80% of variance)



B. AHP (Expert Layer)

Where statistical weighting alone risks undervaluing strategically important but low-variance factors (e.g., affordability in low-income belts), pairwise comparison (Analytic Hierarchy Process) with category level business judgment refines final weights.

Dimension	PCA Weight	AHP-Adjusted Weight
Disease Burden	0.312	0.358
Affordability	0.148	0.202
Access	0.215	0.238
Macroeconomic	0.175	0.180
Total	1.000	1.000

Final weights = statistically grounded, business-calibrated — not arbitrary.

HANDLING GAPS & LOOKING FORWARD



A. Missing Data Treatment

- * District level gap → state average imputation (flagged, not silently filled)
- * Sparse indicators (e.g., DDP in non-reporting states) → top-down allocation using MoSPI's state to district methodology
- * Every imputed value is flagged in our underlying dataset no silent assumptions.



B. Forecasting Future Opportunity

- * 3-year trend extrapolation on Leading Indicators (income growth, urbanization rate) using linear/CAGR based projection
- * Districts re-scored under a "Future MAI" using projected values → identifies movers (districts climbing the rank) vs stable leaders

Illustrative Example
(not from live model):
Gaya, Bihar



TRANSPARENT

Every weight traceable to either statistical output or documented expert rationale nothing arbitrary



AUDITABLE

All imputations flagged assumptions documented in supporting appendix

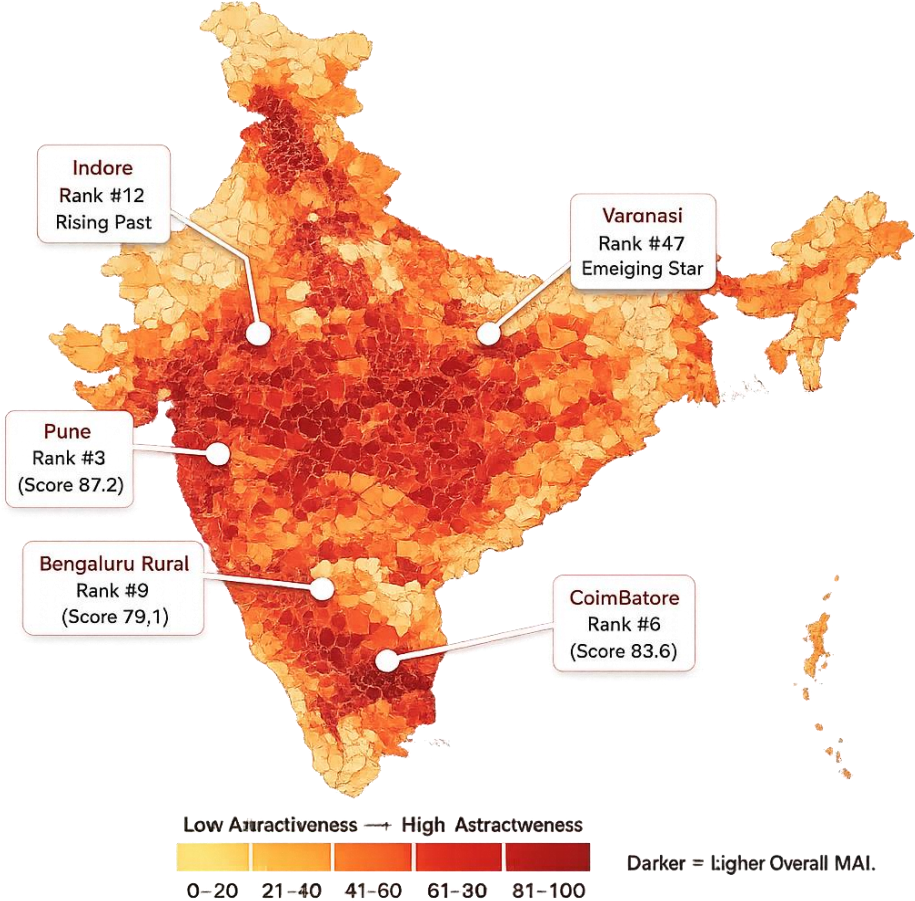


FORWARD COMPATIBLE

Framework reruns automatically as new NFHS/Census cycles release not a one time snapshot.

700+ Districts, Ranked: Where Opportunity Lives

India at a Glance Overall Market Attractiveness Index



TOP 10 DISTRICTS BY OVERALL MAI			
Rank	District	State	Overall MAI
1	Ahmedabad	Gujarat	92.4
2	Mumbai(Suburban)	Maharashtra	90.1
3	Pune	Maharashtra	87.2
4	Bengaluru	Karnataka	85.6
5	Surat	Gujarat	84.3
6	Coimbatore	Tamil Nadu	83.5
7	Hyderabad	Telangana	81.9
8	Chennai	Tamil Nadu	80.4
9	Bengaluru(Rural)	Karnataka	79.1
10	Vadodra	Gujarat	78.6

Also See: Chronic MAI

Acute MAI TOP 10



ESTABLISHED LEADERS

Top-ranked districts cluster around major urban-industrial belts Maharashtra, Gujarat, Tamil Nadu, Karnataka - driven by high infrastructure density and affordability.

Rising Stars

84 districts in Tier-2/3 cities show > 15-point MAI improvement in the Future MAI projection - flagging them as pre-emptive expansion targets before competitors arrive.

Underserved Pockets

136 districts combine high disease burden with low current infrastructure - representing latent demand not yet captured by traditional doctor-density metrics.

From Score to Strategy: Turning Rankings into Deployment Decisions

Sales Force Deployment. Market Prioritization. Territory Planning.



1.SALES FORCE DEPOLOYMENT

Trigger rule:
Districts scoring 75+ on Overall MAI with <1 rep per 50,000 patients → flagged for rep addition

	Current Reps	Recommended Reps
Ahmedabad	12	18
Coimbatore	9	15
Guntur	6	12

Number of reps →

 Right-sizes headcount to real opportunity, not legacy assumptions



2.MARKET PRIORITISATION

Trigger rule:
Tier districts into Priority 1 (Top 15%), Priority 2 (Next 35%), Priority 3 (Remaining) by Overall MAI, cross-checked against therapy specific rank.

Priority 1
(Top 15%)

Priority 2
(Top 30%)

Priority 3
(Remaining)

105 districts

245 districts

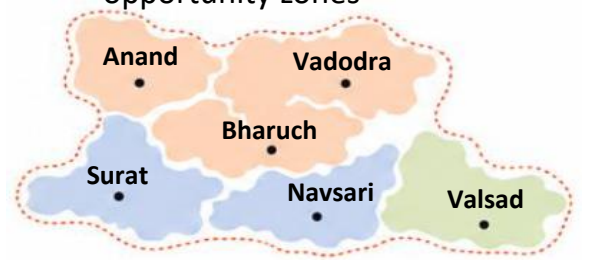
350 districts

 Resources flow to highest-yield markets first - not evenly spread



3.TERRITORY PLANNING

Trigger rule:
Cluster adjacent high MAI districts into single territories to minimize rep travel while maximizing coverage of high opportunity zones



 Balances opportunity density with operational feasibility

WORKED EXAMPLE: FROM SCORE TO ACTION - GUNTUR



THE SCORE

Guntur: Overall MAI 82, Chronic Rank #4, Acute Rank #52



THE READ

High chronic opportunity, low acute opportunity chronic-first territory profile



THE DECISION

Add 2 reps focused on chronic portfolio; deprioritize acute campaign spend here



THE OUTCOME (PROJECTED)

Estimated 18-22% uplift in chronic prescription volume within 12 months* Illustrative projection



HIGH POTENTIAL TODAY

312 districts already scoring 75+ Overall MAI ready for immediate resource reallocation.



FUTURE HIGH GROWTH

186 districts projected to cross the 75+ threshold within 3 years based on Future MAI pre-emptive entry opportunity before competitors.

Where We Drew the Line: Assumptions, Gaps & Honest Limitations

Rigor Means Knowing What You Don't



KEY ASSUMPTIONS

- District boundaries follow Census 2011 delineation; newly formed districts mapped to parent district where data is unavailable.
- Equal data reliability assumed across states, though reporting quality varies by state health system maturity.
- Missing values treated via state-average imputation, assumes intra-state homogeneity where district specific data is absent.
- Future MAI projections assume linear/CAR trend continuation does not account for policy shocks or sudden infrastructure investment.



DATA LIMITATIONS

- Census 2011 is the latest full demographic census 15 year old base, growth adjusted but not a live count.
- District Domestic Product (DDP) coverage varies by state; not all 36 states/UTs currently compile it uniformly.
- NVBDCP/IDSP outbreak data reflects reported cases - likely undercounts true incidence in low surveillance capacity districts.
- Private healthcare sector (large share of Indian care delivery) is underrepresented in public infrastructure datasets.



MODEL LIMITATIONS

- Weights (PCA + AHP) reflect current data patterns may need recalibration as new NFHS/Census cycles release.
- Model does not yet incorporate competitor presence or existing Sun Pharma market share purely opportunity side, not competitive-position-side.
- Mobility & Access Friction Index is a constructed proxy (no official source) directional, not precisely validated.
- Static therapy categorization (Chronic/Acute) some conditions (e.g., certain infections requiring long treatment) may blur the line.

Mitigation Approach : Not Just Flagged, Addressed

Outdated Census base	• Layered with 3-year DDP/urbanization growth trends to approximate current state.
Uneven DDP coverage	• Top-down state-to-district allocation per MoSPI's published methodology.
Constructed proxy (Mobility Index)	• Clearly labeled as directional; excluded from core PCA, added as supplementary +1 layer.
Static Chronic/Acute split	• Categorization reviewed against standard WHO ICD groupings, not ad hoc judgment.



Every assumption here is documented, not hidden -because a defensible framework is one that shows its seams. Full data dictionary, source list, and imputation log available in supporting appendix.

From Framework to Impact: What Happens Next

Turning 700+ District Scores Into a Growth Roadmap



700+

Districts scored across 3 indices



8 + 1

Dimensions + unique Mobility variable



312 → 498

Districts high-potential today → today + future-growth combined

RECOMMENDED PATH FORWARD



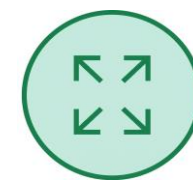
PHASE 1

Neat-Immediate
(0-9 months)



PHASE 2

Neat-Term
(3-12 months)



PHASE 3

Long-Term
(12- months)

- Deploy resources to Priority 1 districts (105 districts, Top 15% Overall MAI)
- Begin chronic-first vs acute-first territory reclassification using divergence analysis

- Validate model against actual Q1-Q2 sales performance in reprioritized districts
- Refine AHP expert weights based on real-world outcome feedback

- Extend framework to additional therapy areas (Oncology, Respiratory, per Slide 9)
- Integrate competitor presence & market share data to move from opportunity-only to opportunity-vs-position scoring
- Refresh with NFHS-6 / new Census cycle when released



Market attractiveness isn't static - and neither should your strategy be. This framework replaces guesswork with granular, forward-looking intelligence - turning 700+ districts into a living, prioritized growth roadmap for Sun Pharma.